



MATHEMATICS METHODS : UNITS 1 & 2, 2023

PM

Test 4 – (10%)

2.2.4-2.2.9, 2.3.2-2.3.17, 2.3.20-2.3.21

Time Allowed 20 minutes	First Name	Surname	Marks
			23 marks

Circle your Teacher's Name:

Mrs Alvaro	Mr Bartoshefski	Mr Buckland	Mr Liu
Mr Lockyer	Mr Luzuk	Mr Mohammadi	
Ms Narendranathan		Mr Ng	
Mr Stephens	Mr Tanday		

Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

- ❖ Calculators: Not Allowed
- ❖ Formula Sheet: Provided
- ❖ Notes: Not Allowed

PART A – CALCULATOR FREE

Question 1 (3 marks)

For an arithmetic series, the first term is 10 and the sum of the first 2 terms is 16.

(a) Show that the common difference is -4.

$$T_1 = 10$$

$$S_2 = T_1 + T_2$$

$$16 = 10 + T_2$$

$$T_2 = 6$$

$$\begin{aligned} d &= T_2 - T_1 \\ &= 6 - 10 \\ &= -4 \end{aligned}$$

OR

$$a + a + d = 16 \checkmark$$

$$10 + 10 + d = 16 \checkmark$$

$$20 + d = 16$$

$$d = -4 \checkmark$$

✓ shows $S_2 = T_1 + T_2$
 ✓ substitutes values
 for S_2 & T_1
 to find T_2
 ✓ Shows $d = -4$.

Question 2 (1, 2, 3 = 6 marks)

- (a) Determine $\frac{dy}{dx}$ for the following, leaving your answer in simplest form, with positive indices:

(i) $y = 3x^4 - 2x + 5$

$$\frac{dy}{dx} = 12x^3 - 2$$

✓ correct derivative

(ii) $y = \sqrt[3]{x} - 5x$

$$y = x^{1/3} - 5x$$

$$\frac{dy}{dx} = \frac{1}{3}x^{-2/3} - 5$$

$$= \frac{1}{x^{2/3}} - 5$$

$$\text{OR } \frac{1}{\sqrt[3]{x^2}} - 5$$

✓ correct derivate
✓ expresses with positive indices

- (b) Find the value of the gradient of $y = \frac{8x^4 - 3x^2}{x^3}$ at $x = 3$.

$$y = 8x - \frac{3}{x}$$

$$y = 8x - 3x^{-1}$$

$$\frac{dy}{dx} = 8 + 3x^{-2}$$

$$= 8 + \frac{3}{x^2}$$

$$\left. \frac{dy}{dx} \right|_{x=3} = 8 + \frac{3}{9}$$

$$= 8\frac{1}{3}$$

✓ simplifies y

✓ correct derivate

✓ correct value at $x=3$ (FT)

Question 3 (4 marks)

Differentiate $f(x) = 2x^2 - x$ by first principles.

$$\begin{aligned} & \lim_{h \rightarrow 0} \frac{2(x+h)^2 - (x+h) - (2x^2 - x)}{h} && \checkmark \text{establish equation} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{2x^2} + 4xh + 2h^2 - \cancel{x} - h - \cancel{2x^2} + \cancel{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 - h}{h} && \checkmark \text{expands and simplifies} \\ &= \lim_{h \rightarrow 0} \frac{\cancel{h}(4x + 2h - 1)}{\cancel{h}} && \checkmark \text{factors out } h \text{ and cancels} \\ &= 4x - 1 + \lim_{h \rightarrow 0} 2h \\ &= 4x - 1 && \checkmark \text{works limit and simplifies to final answer.} \end{aligned}$$

Question 4 (5, 3, 2 = 10 marks)

A function has the following rule $y = 2x^3 - 6x^2 + 1$ for $-1 \leq x \leq 4$.

- (a) Determine, using calculus, all the stationary points and their nature, using the sign test, for the function given above.

$$\frac{dy}{dx} = 6x^2 - 12x$$

✓ finds derivative and assigns to zero

$$0 = 6x^2 - 12x$$

$$0 = 6x(x-2)$$

$$x = 0, 2$$

✓ finds two values for x

$$\begin{aligned} y &= 2(2)^3 - 6(2)^2 + 1 \\ &= 16 - 24 + 1 \\ &= -7 \end{aligned}$$

$$x = -1 \quad x = 0 \quad x = 1$$

$$\frac{dy}{dx} = +ve \quad \frac{dy}{dx} = 0 \quad \frac{dy}{dx} = -ve$$

$$x = 1 \quad x = 2 \quad x = 3$$

$$\frac{dy}{dx} = -ve \quad \frac{dy}{dx} = 0 \quad \frac{dy}{dx} = +ve$$

Local Maximum at (0, 1)

Local Minimum at (2, -7)

✓ use sign test to prove max at (0, 1)

✓ use sign test to prove min at (2, -7)

✓ writes max/min as coordinates.

- (b) Determine the global maxima and minima over the given interval.

Test end points. at $x = -1$ $y = 2(-1)^3 - 6(-1)^2 + 1$

$$\begin{aligned} &= -2 - 6 + 1 \\ y &= -5 \end{aligned}$$

✓ Tests $x = -1$

$$\begin{aligned} \text{at } x = 4 \quad y &= 2(4)^3 - 6(4)^2 + 1 \\ &= 128 - 96 + 1 \\ &= 33 \end{aligned}$$

✓ Tests $x = 4$

$$\text{Global maxima} = 33 \quad \text{Global minima} = -7$$

✓ global max and min correct

- (c) Discuss the behaviour of the function as $x \rightarrow \pm\infty$.

$$\text{As } x \rightarrow \infty \quad y \rightarrow +\infty \quad \checkmark$$

$$\text{As } x \rightarrow -\infty \quad y \rightarrow -\infty \quad \checkmark$$

writes correct global max & global min

	MATHEMATICS METHODS : UNITS 1 & 2, 2023 Test 4 – (10%) 2.2.4-2.2.9, 2.3.1-2.3.17, 2.3.20-2.3.21		PM
Time Allowed 30 minutes	First Name Surname <i>Solutions</i>	Marks 32 marks	

Circle your Teacher's Name:

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Mr Lockyer	Mr Luzuk	Mr Mohammadi	
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Assessment Conditions: (N.B. Sufficient working out must be shown to gain full marks)

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PART A – CALCULATOR ASSUMED

Question 5 (5, 2 = 7 marks)

If $(t + 1)$, $(1 - t)$ and $(2 - 5t)$ are the first three terms of an infinite, **converging** geometric sequence, determine:

(a) the value(s) of t and the common ratio.

$$\frac{2-5t}{1-t} = \frac{1-t}{t+1} \quad \checkmark \text{ sets up equation}$$

$$t = -\frac{1}{2}, \frac{1}{3} \quad \checkmark \text{ finds two solutions for } t \text{ (CLASSPAD SOLVE)}$$

$$\begin{aligned} \text{When } t = -\frac{1}{2} \\ r &= \frac{1 - (-\frac{1}{2})}{-\frac{1}{2} + 1} \\ &= 3 \end{aligned}$$

$\checkmark$ finds r for $t = -\frac{1}{2}$

$$\begin{aligned} \text{When } t = \frac{1}{3} \\ r &= \frac{1 - \frac{1}{3}}{\frac{1}{3} + 1} \\ &= \frac{1}{2} \end{aligned}$$

$\checkmark$ finds r for $t = \frac{1}{3}$

$\checkmark$ reject $r = 3$ as sequence is converging.

(b) the sum to infinity.

use $r = \frac{1}{2}$ and $a = 1\frac{1}{3}$ $\checkmark$ selects appropriate r and a .

$$S_{\infty} = \frac{1\frac{1}{3}}{1 - \frac{1}{2}}$$

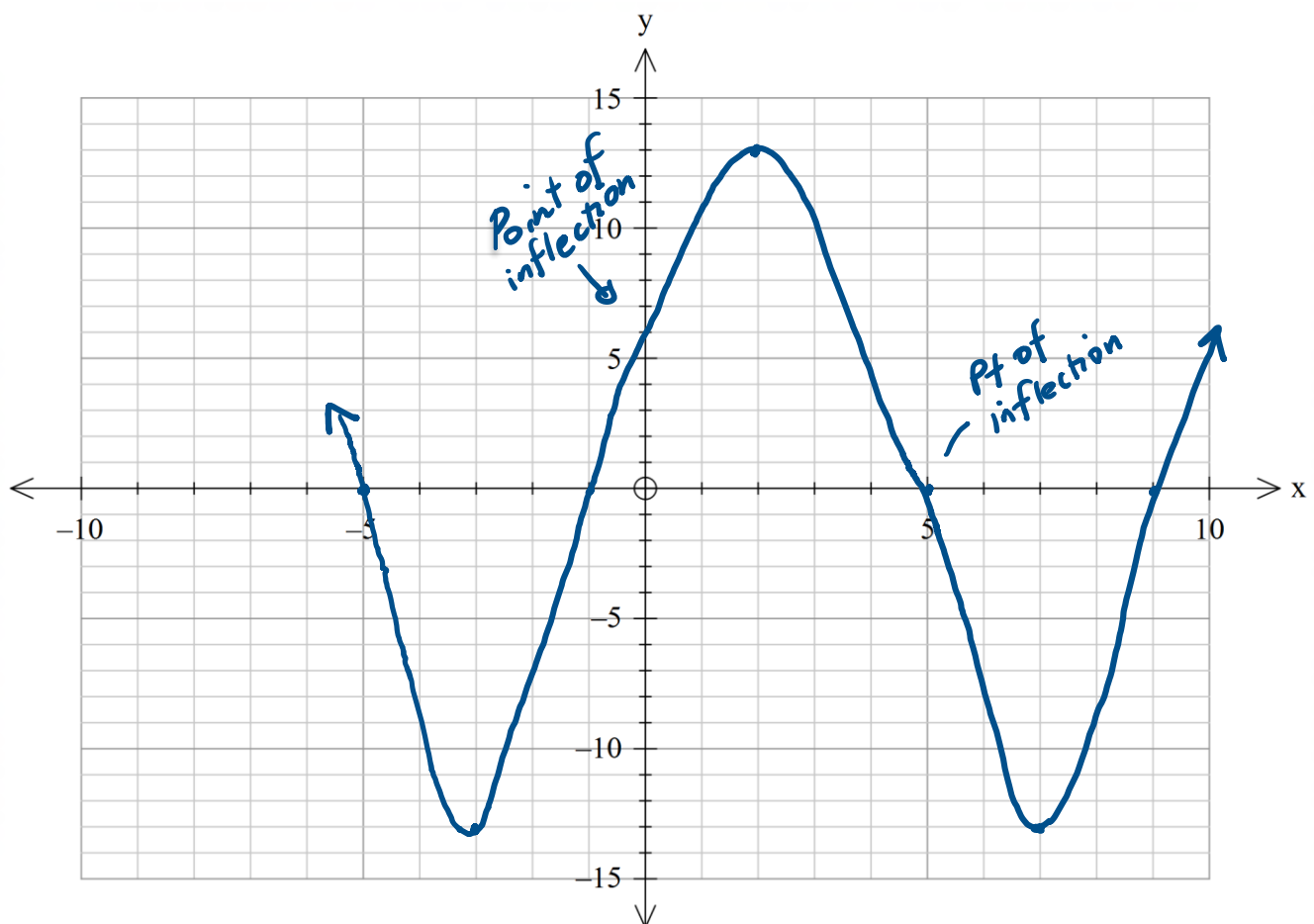
$$= \frac{8}{3} \text{ or } 2\frac{2}{3} \quad \checkmark \text{ finds } S_{\infty}$$

Question 6 (4 marks)

A function has the following characteristics:

- The function has stationary points at $x = -3$, $x = 2$, and $x = 7$.
- The function has roots at $x = -5$, $x = -1$, $x = 5$, $x = 9$.
- The function has a local maximum at $x = 2$ and local minimums at $x = 7$ and $x = -3$.
- $f'(x) > 0$ for $-3 < x < 2$ and $x < 7$.
- $f'(x) < 0$ for $x < -3$ and $2 < x < 7$.
- The function has a y-intercept at $(0,6)$.
- The function changes concavity at $x = 0$ and $x = 5$.

Graph an example of the function on the axes provided below.



✓ roots in correct position
✓ correct max x-value
✓ correct mins x-values
✓ pts of inflection indicated
in correct positions with
appropriate gradient.

Question 7 (2, 2 = 4 marks)

Determine the coordinates on the graph of $f(x) = x^3 - 3x^2 + 3x + 4$ where the tangent

(a) has a gradient of 12.

$$f'(x) = 3x^2 - 6x + 3$$

$$12 = 3x^2 - 6x + 3$$

$$x = -1, 3$$

✓ finds derivative
and assigns to 12.

$$(-1, -3) \quad (3, 13)$$

✓ two correct points as
coordinates

(b) is parallel to the line $3x - y = -4$.

$$y = 3x + 4$$

$$m = 3$$

$$3 = 3x^2 - 6x + 3$$

$$x = 0, 2$$

✓ assigns derivative
equal to 3

$$(0, 4) \quad (2, 6)$$

✓ two correct points as
coordinates

Rounding question

Question 8 (2, 2, 1, 3, 3 = 11 marks)

(-1) overall if not rounded to nearest cent.

Indy accepts employment with a first year salary of \$52 000. In each of the following years her annual salary is increased by \$2575 per year.

Skye accepts employment with a first year salary of \$52 000. In each of the following years her annual salary is increased by 4%.

- (a) State the recursive formula for Indy's annual salary.

$$T_{n+1} = T_n + 2575 \quad T_1 = \$52000 \quad \text{— salary for first year.}$$

- (b) State the general (explicit) formula for Skye's annual salary during the n^{th} year.

$$T_n = 52000 (1.04)^{n-1}$$

OR $T_n = 50000 (1.04)^n$

✓ uses 1.04
✓ writes correct general form (either answer)

- (c) Determine Skye's annual salary during the 10th year.

$$T_{10} = 52000 (1.04)^9$$
$$= \$74\,012.21$$

✓ Correct \$ amount to nearest cent.

- (d) During which year does Skye earn more than Indy for the first time? Show evidence to justify your answer.

$$52000 + 2575(n-1) = 52000 (1.04)^{n-1}$$
$$n = 13^{\text{th}} \text{ year (using SEQUENCE APP)} \quad \text{✓ Finds correct } n.$$

In 12th year Indy = \$80325 Skye = \$80052 ✓ shows values for 12th year

In 13th year Indy = \$82900 Skye = \$83254 ✓ shows values for 13th year.

- (e) By what amount does the total amount earned by Skye exceed that earned by Indy at the end of the twentieth year?

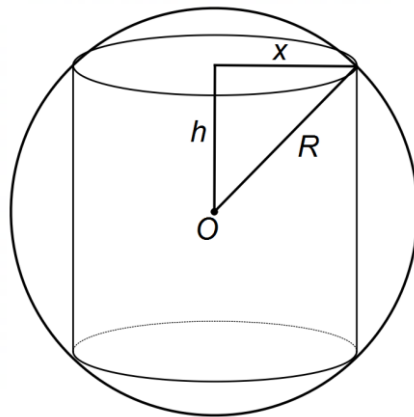
$$\begin{aligned} \text{Skye} \Rightarrow S_{20} &= \frac{52000(1.04^{20} - 1)}{(1.04 - 1)} \\ &= \$1548460.09 \quad \checkmark \text{ correct Sum for Skye} \end{aligned}$$

$$\begin{aligned} \text{Indy} \Rightarrow S_{20} &= \frac{20}{2} [2(52000) + (20-1)(2575)] \\ &= \$1529250 \quad \checkmark \text{ correct Sum for Indy} \end{aligned}$$

$$\text{Difference} = \$19210.09. \quad \checkmark \text{ correct difference}$$

Question 9 (3, 3 = 6 marks)

A cylinder, of radius x metres and height $2h$ metres, is to be inscribed in a sphere of constant radius R (m) centred at O as shown below.



- (a) Show that the volume of the cylinder is given by $V = 2\pi h(R^2 - h^2)$.

$x^2 = R^2 - h^2$ ✓ shows how x^2 is found

$V(\text{cylinder}) = \pi x^2(2h)$ ✓ expresses volume of cylinder in terms of x^2 and h
 $= 2\pi h x^2$

$= 2\pi h(R^2 - h^2)$ ✓ subs for x^2 and completes.

- (b) Use **calculus techniques** to show that the cylinder has a maximum volume

when $h = \frac{\sqrt{3}R}{3}$.

$V = 2\pi h(R^2 - h^2)$
 $= 2\pi R^2 h - 2\pi h^3$

$\frac{dV}{dh} = 2\pi R^2 - 6\pi h^2$

$0 = 2\pi R^2 - 6\pi h^2$ ✓ differentiates with respect to h and equates to 0.

$6\pi h^2 = 2\pi R^2$

$h^2 = \frac{2\pi R^2}{6\pi}$ ✓ isolates h^2 .

$h^2 = \frac{R^2}{3}$

$h = \frac{R}{\sqrt{3}} = \frac{\sqrt{3}R}{3}$ ✓ simplifies to rationalised required value